

Sets



Outline

- Informal introduction to set theory

 - Informal definition of sets

- Number sets

- Relations between sets

 - Empty set, cardinality and power set

- Operations on sets

 - Complementary sets and disjoint sets

 - Venn diagrams

 - Cartesian product

 - Properties of operations on sets

 - Properties of the cardinality of sets

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 - Russell's paradox

Informal introduction to set theory



Introduction: Set theory

- ▶ In this section we informally introduce the concepts of **set**, **element** and **membership**
- ▶ We will see how a set is defined: **by listing** or **by a property**
- ▶ Sets are the basic language of all of mathematics and of this course

The three key concepts (1/2)

Set theory is based on the following three key concepts:

1. **Sets:** the notion of set is generally taken as **primitive** (that is, not reducible to more elementary concepts). The words *collection*, *class*, *aggregate*, *family* are used as synonyms of set.
2. **Elements:** a set is determined by its *elements*, in the sense that a set is defined when we have a **criterion** to establish whether a given element is or is not an element of this set. There are sets with a *finite number of elements* (for example the set of students of a university) and sets with an *infinite number of elements* (for example the set of points of the plane).

Example

Sets.

For example, we have the set of points of a plane, the set of students of a university, or the set of stars of a galaxy.

The three key concepts (2/2)

Sets are usually denoted by capital letters. The elements of a set are usually denoted by lowercase letters.

3. **Membership:** the concept of *membership* links elements to sets. When an object is an element of a set, we say that the element *belongs* to the set. To indicate that an element x belongs to a set A we write:

$$x \in A$$

Definition by listing

- ▶ A first way to define sets is **definition by listing** (roster notation).

A set can be defined **by listing**, that is, by listing the elements that belong to it between curly brackets. This technique assumes that the set has a finite number of elements.

Example

Definition by listing. The notation:

$$A = \{a, b, c\}$$

means that the set A has as elements the three letters a , b and c .
For example, a belongs to A , that is, $a \in A$.

Definition by a property

- ▶ A second way to define sets is **definition by a property**.

A set can be defined **by a property** as follows:

$$A = \{ x \in U : p(x) \text{ is true} \}$$

where $p(x)$ is the property that the element x of the set U must have in order to belong to the set A . This technique can be used to define sets with a finite or even an infinite number of elements.

Example

Definition by a property. Consider for example the set of letters of the Latin alphabet:

$$U = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$$

Using for example the property $p(x)$ defined as “ x is a vowel” we can define the following set of vowels:

$$A = \underbrace{\{x \in U : x \text{ is a vowel}\}}_{\{a, e, i, o, u\}}$$

Universal set

- ▶ Note that to define a set A by a property we need a set U to which all the elements of the set A we want to define belong. The set U plays the role of the **universal set**.
- ▶ It is important that the property $p(x)$ we use makes sense for every x of the set U (universal set), and therefore is true or false (without ambiguity of meaning) for every particular $x \in U$; the set A will then consist of exactly those x belonging to U for which the property $p(x)$ is true.

Beware of contradictions

One must be careful when defining sets, since contradictions may arise. There is a formal definition of the concept of set, developed to avoid contradictions, but it is beyond the scope of this course.

- ▶ For example, the set of all sets that do not contain themselves is not a set in the formal definition of sets. Admitting this set would generate the contradiction: “the set of all sets that do not belong to themselves belongs to itself if and only if it does not belong to itself” (**Russell's paradox** – section Further topics).

Summary

- ▶ Key concepts: **set** (primitive notion), **element**, **membership** ($x \in A$)
- ▶ Sets with capital letters, elements with lowercase letters
- ▶ Definition **by listing**: $A = \{a, b, c\}$ (finite sets)
- ▶ Definition **by a property**: $A = \{x \in U : p(x) \text{ is true}\}$, with U the universal set
- ▶ A naive definition can lead to contradictions (Russell's paradox)

Number sets



Introduction: Number sets

- ▶ In this section we introduce **positional numeral systems** and the decimal expansion
- ▶ We give an informal definition of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ through decimal expansions
- ▶ We will see why $0,\overline{9} = 1$ and examples of real numbers that are not rational

Definition 1: numeral system

*A **positional numeral system** is a way of encoding numbers using a **sequence of digits**, where each digit contributes differently to the number depending on its position. The number of distinct digits is the **base** of the system.*

The **decimal expansion** of a number is a numerical representation in base 10; it expresses a number as a sum of powers of 10, and it can be finite or infinite depending on the type of number.

Informal definition of the main number sets (1/2)

1. We denote by $\mathbb{N}$ the set of **natural numbers**, that is, the set of numbers that can be written as **decimal expansions without decimal point and without sign**. We will use the informal notation:

$$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$$

2. We denote by $\mathbb{Z}$ the set of **integers**, that is, the set of numbers that can be written as **decimal expansions without decimal point and with a sign**. We will use the informal notation:

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \pm 4, \dots\}$$

Informal definition of the main number sets (2/2)

3. We denote by $\mathbb{Q}$ the set of **rational numbers**, that is, the set of numbers that can be written as **finite or infinite periodic decimal expansions**. In other words, it is the set of numbers that can be written as a fraction $\frac{p}{q}$ where p is an integer and q is a natural number different from zero.

Example

Rational numbers.

- ▶ For example, with $p = 2$ and $q = 5$ we have the fraction $\frac{2}{5}$, whose decimal expansion is 0.4.
- ▶ For example, with $p = 4$ and $q = 10$ we have the fraction $\frac{4}{10}$, whose decimal expansion is again 0.4. Note that a rational number can be written with more than one fraction.
- ▶ For example, with $p = 13$ and $q = 30$ we have the fraction $\frac{13}{30}$, whose decimal expansion is $0.4\bar{3} = 0.43333\dots$

However, we can represent every rational number different from 0 by a single fraction $\frac{p}{q}$ by choosing $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ coprime (that is, relatively prime: p and q are not both divisible by the same integer greater than 1).

The number $0,\overline{9}$

Observation 1

$$0,\overline{9} = 1$$

- There are different proofs of this remark, based on different mathematical techniques.

Proof (1/3)

1. A simple proof follows directly from the definition of 1 divided by 3; indeed, we have:

$$\frac{1}{3} = 0,\overline{3}$$

$$\frac{1}{3} \cdot 3 = 0,\overline{3} \cdot 3$$

$$1 = 0,\overline{9}$$

Proof (2/3)

2. Using algebraic arguments we can write:

$$x = 0.999 \dots$$

$$10x = 9.999 \dots \quad \text{multiplying by 10}$$

$$10x = 9 + 0.999 \dots \quad \text{separating the integer part} \\ \text{from the fractional part}$$

$$10x = 9 + x \quad \text{by definition of } x$$

$$9x = 9 \quad \text{subtracting } x$$

$$x = 1 \quad \text{dividing by 9}$$

Proof (3/3)

3. A proof by contradiction is the following:

$$0,\overline{9} \neq 1$$

$$0,\overline{9} \cdot 9 \neq 1 \cdot 9$$

$$0,\overline{9} \cdot 9 + 0,\overline{9} \neq 1 \cdot 9 + 0,\overline{9}$$

$$0,\overline{9} \cdot 9 + 0,\overline{9} \neq 9,\overline{9}$$

$$0,\overline{9} \cdot (9 + 1) \neq 9,\overline{9}$$

$$0,\overline{9} \cdot (10) \neq 9,\overline{9}$$

$$9,\overline{9} \neq 9,\overline{9} \quad \text{contradiction!}$$



The real numbers

4. We denote by $\mathbb{R}$ the set of **real numbers**, that is, the set of numbers identified with finite or infinite decimal expansions, periodic or non-periodic.
- ▶ There are formal definitions of the sets of natural, integer and rational numbers that are beyond the scope of this course. Later we will give a formal definition of the real numbers.

Example

Real numbers.

- ▶ Consider for example the number

$$0,10110111011110\dots$$

obtained by putting after the decimal point one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 ... and so on. The string of nonzero digits after the decimal point is neither finite nor periodic: therefore this number is real but not rational.

- ▶ Other examples of real but not rational numbers are $\sqrt{2}$ and $\sqrt{3}$, or π and Euler's number e (Napier's constant), which have infinite non-periodic decimal expansions and are therefore real but not rational numbers.

Summary

- ▶ **Positional numeral system:** sequence of digits whose contribution depends on the position; base 10
- ▶ $\mathbb{N}$: expansions without decimal point and without sign; $\mathbb{Z}$: without decimal point and with sign
- ▶ $\mathbb{Q}$: finite or infinite periodic expansions, fractions $\frac{p}{q}$
- ▶ $0,\overline{9} = 1$
- ▶ $\mathbb{R}$: arbitrary decimal expansions (e.g. $\sqrt{2}$, π , e are not rational)

Relations between sets



Introduction: Relations between sets

- ▶ In this section we define **equality** and **inclusion** between sets
- ▶ We will learn not to confuse “belongs to” ($\in$) and “is contained in” ($\subseteq$)
- ▶ We will introduce the empty set, cardinality and the power set

Equality and inclusion

1. **Equality:** two sets A and B are equal when they have the same elements. We write

$$A = B$$

and this means that every element that belongs to A also belongs to B and every element that belongs to B also belongs to A .

2. **Inclusion:** it may happen that only one of the two requirements expressed by the equality relation holds. For example, if we only know that every element of A is also an element of B , we can say that A is contained in B . We write:

$$A \subseteq B \quad \text{or} \quad A \subset B$$

and we read “ A is contained in B ” or “ A is a **subset** of B ”. If we state that $A \subseteq B$, we do not exclude that $A = B$.

Membership and inclusion

Saying that $A = B$ is equivalent to saying that $A \subseteq B$ and $B \subseteq A$.

Be careful not to confuse “belongs to” and “is contained in” (in symbols, $\in$ and $\subseteq$). In a sense, they are two ways of indicating that “something is inside something else”, but they have a fundamental logical difference:

- ▶ a set is contained in another set;
- ▶ an element belongs to a set.

Example

“Belongs to” vs “is contained in”. For example:

$$3 \in \{1, 3, 4\}; \quad \{3\} \subseteq \{1, 3, 4\}; \quad \{1, 4\} \subseteq \{1, 3, 4\}.$$

In particular, do not confuse 3 (which is a number) with $\{3\}$, which is the set that contains the number 3 as its only element.

Sometimes we consider sets whose elements are other sets. In this case too, the symbols $\in$ and $\subseteq$ are not interchangeable, but must be used correctly.

Example

“Belongs to” vs “is contained in”. For example, if we define the set

$$A = \left\{ \{1\}, \{2\}, \{1, 2\} \right\},$$

then it is correct to state that $\{2\} \in A$, because $\{2\}$ is a set that plays the role of an element of A , whereas it would be incorrect to say that $\{2\} \subseteq A$, because this would mean that $2 \in A$, while A has $\{2\}$ as an element but not 2.

Instead, the subset of A containing only the element $\{2\}$ is denoted by $\{\{2\}\}$ and we have $\{\{2\}\} \subset A$.

Empty set

Definition 2: of empty set

*The **empty set** is the set that contains no elements. It is denoted by $\emptyset$.*

Observation 2

Given a set A , we have:

$$\emptyset \subseteq A$$

Proof

To prove it, we should show that every element that belongs to $\emptyset$ also belongs to A ; but no element belongs to $\emptyset$, so the thesis holds. □

Cardinality and power set

Definition 3: of cardinality of a set

*The number of elements of a set A is the **cardinality** of the set. It is denoted by $|A|$.*

Definition 4: of power set

*Given a set A , the set whose elements are all the subsets of A is called the **power set** of A and is denoted by the symbol $\mathcal{P}(A)$.*

- ▶ Every set A has two trivial subsets, namely A itself and the empty set $\emptyset$ (they could coincide, if A is empty).

Example

Power set. For example, given

$$A = \{1, 2, 3\},$$

then

$$\mathcal{P}(A) = \left\{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \right\}$$

Cardinality of the power set

Observation 3

Given a set A with n elements, the power set $\mathcal{P}(A)$ has 2^n elements:

$$|\mathcal{P}(A)| = 2^n$$

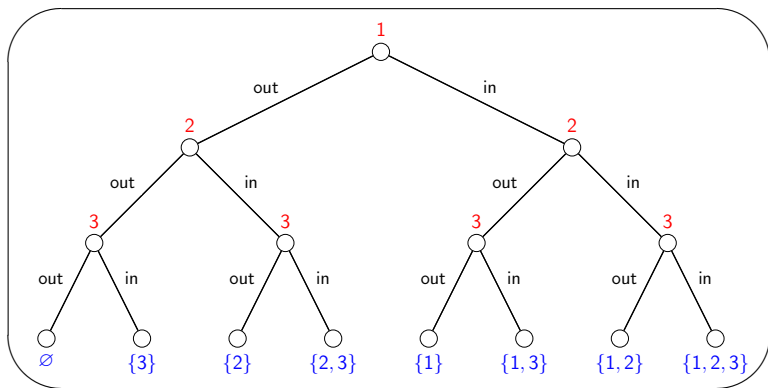
Proof

For each of the elements of A , the subsets of A may or may not contain that element.

Hence we must make a choice between two options n times. The total number of possible subsets is therefore 2^n . □

Example

Construction of the power set with a binary tree. Given the set $A = \{1, 2, 3\}$ with $n = 3$ elements, the cardinality of its power set is $2^3 = 8$. The construction of $\mathcal{P}(A)$ can be visualized through the following **binary tree** (an undirected, connected and acyclic graph) in which at each level we decide whether or not to include the object in the subset:



Summary

- ▶ $A = B$: same elements; equivalent to $A \subseteq B$ and $B \subseteq A$
- ▶ $A \subseteq B$: every element of A is an element of B (subset)
- ▶ $\in$ links elements and sets, $\subseteq$ links sets and sets
- ▶ $\emptyset$ is contained in every set; $|A|$ is the cardinality
- ▶ $\mathcal{P}(A)$: power set, $|\mathcal{P}(A)| = 2^n$ if $|A| = n$



Operations on sets

Introduction: Operations on sets

- ▶ In this section we define **intersection**, **union**, **difference**, **complement** and **Cartesian product**
- ▶ We will represent them graphically with **Venn diagrams**
- ▶ We will study their properties (commutative, associative, distributive, De Morgan's laws) and the properties of cardinality

Intersection and union

Definition 5: of intersection of sets

The intersection of two sets $A, B \subseteq U$ is the set defined by:

$$A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$$

It is the set of elements that belong both to the first and to the second set.

Definition 6: of union of sets

*The **union** of two sets $A, B \subseteq U$ is the set defined by:*

$$A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$$

It is the set of elements that belong to the first or to the second set, where “or” is meant in the non-exclusive sense (the set of elements that belong to A or to B or to both).

Definition 7: of difference of sets

The **difference** of two sets $A, B \subseteq U$ is the set defined by:

$$A \setminus B = \{x \in A : x \notin B\}$$

It is the set of elements that belong to the first but not to the second set. The symbol “ $\setminus$ ” can also be written “ $-$ ” by analogy with arithmetic subtraction.

Example

Universal sets. For example, in questions of arithmetic we could have $U = \mathbb{N}$, while in questions of analysis we could have $U = \mathbb{R}$.

Definition 8: of set complementation and complementary sets

The **complementation** of a set $A \subseteq U$ is the set defined by:

$$\overline{A} = \{x \in U : x \notin A\}$$

This set is called the **complement** (complementary set) of A with respect to U

- For every set $A \subseteq U$, where U is the universal set, we have the following relations.

Properties of the complement

$$\overline{\overline{A}} = A$$

$$A \cap \emptyset = \emptyset$$

$$\overline{U} = \emptyset$$

$$A \cap \overline{A} = \emptyset$$

$$A \cap U = A$$

$$\overline{\emptyset} = U$$

$$A \cup \overline{A} = U$$

$$A \cup \emptyset = A$$

$$A \cup U = U$$

Definition 9: of disjoint sets

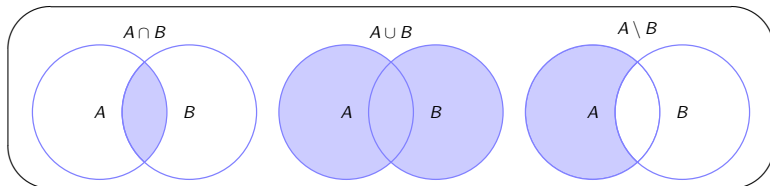
Two sets A and B are **disjoint** if they have no elements in common:

$$A \cap B = \emptyset$$

Venn diagrams

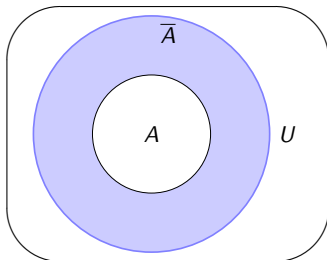
Venn diagrams are graphical representations in which sets are represented as regions of the plane.

- Venn diagrams of intersection, union and difference:



Venn diagrams

- Venn diagram of the complement of a set:



Cartesian product

There is another operation on sets, which can be performed on any two sets (i.e., two sets not necessarily contained in the same universal set):

Definition 10: of Cartesian product

*Given two (not necessarily distinct) sets A and B , the set consisting of all ordered pairs (a, b) , with $a \in A$ and $b \in B$, is called the **Cartesian product** of A and B and is denoted by the symbol $A \times B$.*

When A and B are two finite sets, the cardinality of their Cartesian product is:

$$|A \times B| = |A| \cdot |B|.$$

Example

Cartesian product.

$$\{a, b\} \times \{a, b, c\} = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c)\}$$

$$|\{a, b\}| = 2, \quad |\{a, b, c\}| = 3, \quad |\{a, b\} \times \{a, b, c\}| = 2 \cdot 3 = 6$$

- ▶ A typical use of the Cartesian product is $\mathbb{R} \times \mathbb{R}$, which is abbreviated by the symbol $\mathbb{R}^2$ and denotes the set of ordered pairs of real numbers.
- ▶ Similarly, $\mathbb{R}^n$ (abbreviation of the Cartesian product of n sets equal to $\mathbb{R}$) is the set of *ordered n -tuples of real numbers*

$$\mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}, \quad i \in \{1, 2, \dots, n\} \}$$

Properties of intersection

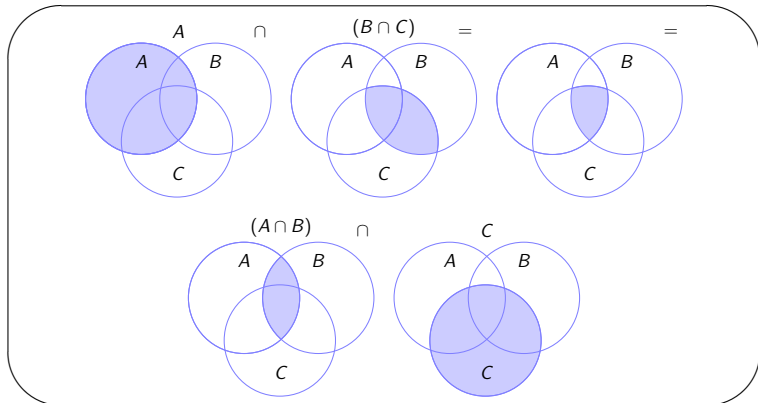
Observation 4: properties of intersection

Given three sets A , B and C , intersection has the following properties:

- ▶ **Commutative:** $A \cap B = B \cap A$
- ▶ **Associative:** $A \cap (B \cap C) = (A \cap B) \cap C$
- ▶ **Idempotence:** $A \cap A = A$

Proof

The graphical proof of the associative property of intersection is the following:



Properties of union

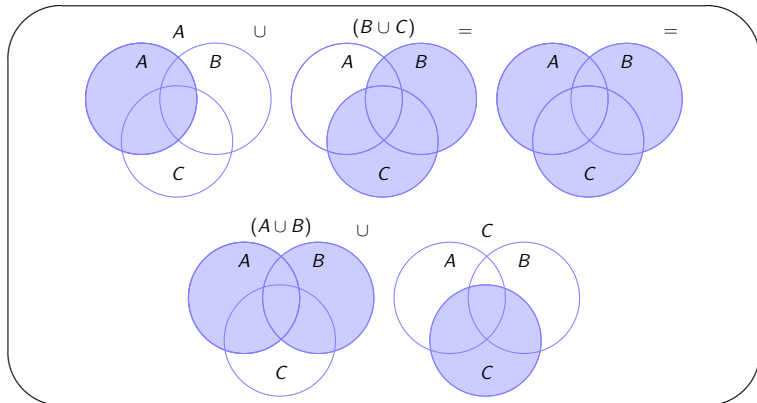
Observation 5: properties of union

Given three sets A , B and C , union has the following properties:

- ▶ **Commutative:** $A \cup B = B \cup A$
- ▶ **Associative:** $A \cup (B \cup C) = (A \cup B) \cup C$
- ▶ **Idempotence:** $A \cup A = A$

Proof

The graphical proof of the associative property of union is the following:



Distributive properties

Observation 6: distributive properties (linking union and intersection)

Given three sets A , B and C we have:

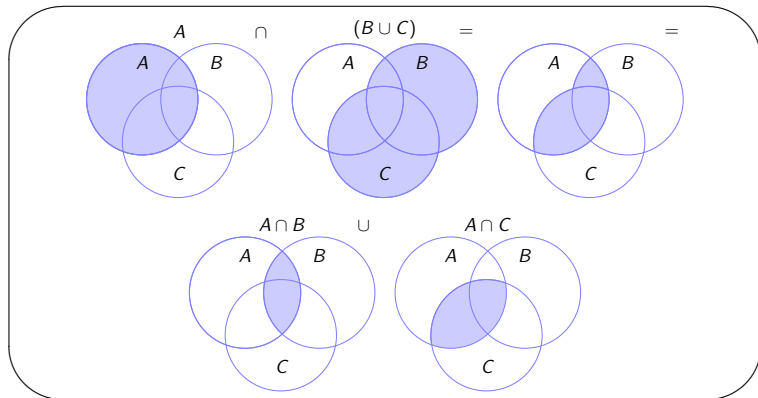
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The distributive properties link union and intersection to each other.

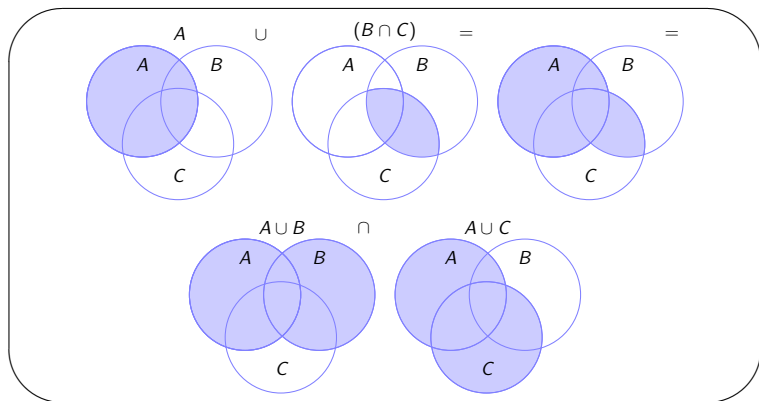
Proof (1/2)

The graphical proof of the first property is the following:



Proof (2/2)

The graphical proof of the second property is the following:



De Morgan's laws

Proposition 1: De Morgan's laws (first version)

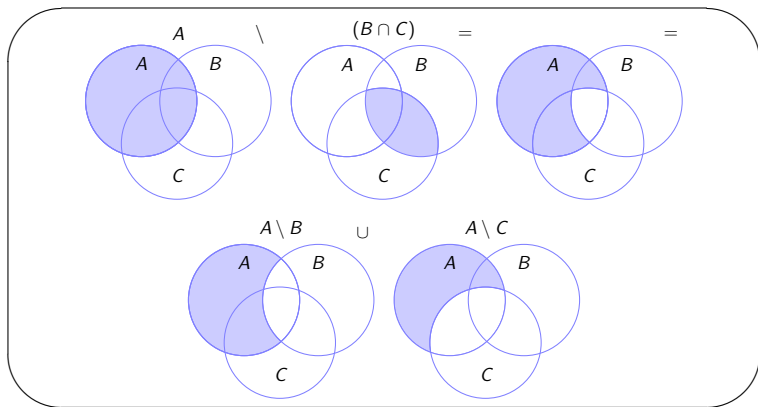
Given three sets A , B and C we have:

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

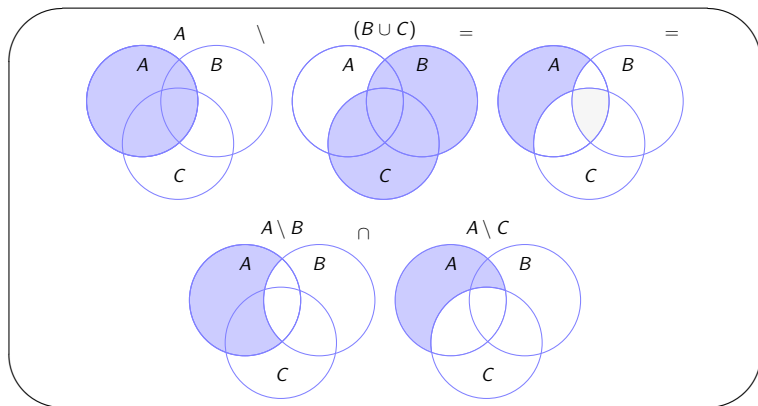
Proof (1/2)

The graphical proof of the first property is the following:



Proof (2/2)

The graphical proof of the second property is the following:



De Morgan's laws

Proposition 2: De Morgan's laws (second version)

Given the sets $B, C \subseteq U$, we have

$$\overline{B \cap C} = \overline{B} \cup \overline{C}$$

$$\overline{B \cup C} = \overline{B} \cap \overline{C}$$

Proof

These laws can be derived by setting A equal to U in the previous De Morgan's laws, as follows:

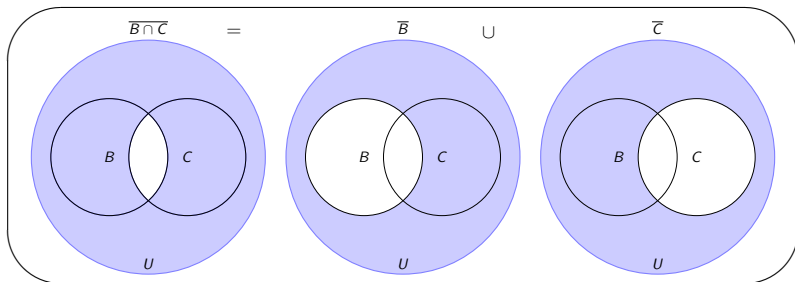
$$\underbrace{U \setminus (B \cap C)}_{= \overline{B \cap C}} = \underbrace{(U \setminus B)}_{= \overline{B}} \cup \underbrace{(U \setminus C)}_{= \overline{C}}$$

$$\underbrace{U \setminus (B \cup C)}_{= \overline{B \cup C}} = \underbrace{(U \setminus B)}_{= \overline{B}} \cap \underbrace{(U \setminus C)}_{= \overline{C}}$$



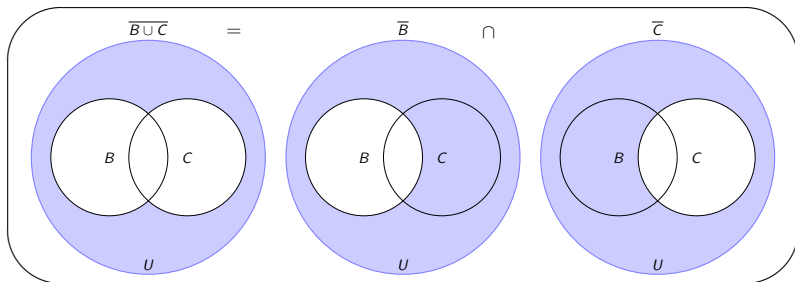
Proof (graphical) (1/2)

The graphical proof of the first property is the following:



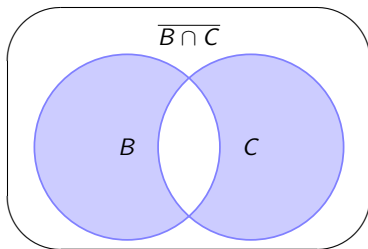
Proof (graphical) (2/2)

The graphical proof of the second property is the following:



Exclusive union

- ▶ The set of elements that belong to a set B or to a set C but not to both (meaning “or” in the exclusive sense) is obtained by setting $U = B \cup C$ and taking the complement of the intersection:



Inclusion, union and intersection

Relation between inclusion and the operations of union and intersection:

$A \subseteq B$ if and only if $A \cap B = A$ and $A \cup B = B$.

Properties of the cardinality of sets

- For any two finite sets A and B , we have

$$|A \cup B| = |A| + |B| - |A \cap B|$$

from which we conclude that

$$|A \cup B| \leq |A| + |B|$$

- If A and B are disjoint then

$$|A \cap B| = 0 \quad \text{and hence} \quad |A \cup B| = |A| + |B|$$

- If $A \subsetneq B$, then $|A| < |B|$

Summary (1/2)

- ▶ $A \cap B$: common elements; $A \cup B$: elements of at least one of the two; $A \setminus B$: elements of A not in B
- ▶ $\overline{A}$: complement with respect to the universal set U ; disjoint if $A \cap B = \emptyset$
- ▶ Venn diagrams: graphical representation of the operations
- ▶ $A \times B$: ordered pairs, $|A \times B| = |A| \cdot |B|$

Summary (2/2)

- ▶ $\cap$ and $\cup$: commutative, associative, idempotent
- ▶ Distributive properties between $\cap$ and $\cup$
- ▶ De Morgan's laws: $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$,
 $\overline{B \cap C} = \overline{B} \cup \overline{C}$, ...
- ▶ $A \subseteq B \Leftrightarrow A \cap B = A$ and $A \cup B = B$
- ▶ $|A \cup B| = |A| + |B| - |A \cap B|$

Further topics

Introduction: Further topics

- ▶ In this section we discuss **Russell's paradox**, which shows the limits of a naive definition of set
- ▶ We mention the Zermelo-Fraenkel axioms and prove that no set is an element of itself
- ▶ Finally we see a further proof of $0, \bar{9} = 1$ based on the Archimedean property

Russell's paradox

“can a set be an element of itself or not?”

- ▶ For example, the set of all books in a library is not an element of itself (a set of books is not a book). On the other hand, the set of all sets with more than 20 elements is an element of itself.
- ▶ Following this reasoning, two categories of sets can be defined:
 1. sets that are not elements of themselves
 2. sets that are elements of themselves

If we consider the set of all sets that are not elements of themselves, is it an element of itself or not?

Russell's paradox

Let us call this set S ; two hypotheses can be made:

- 1 If we suppose $S \in S$, then S contains itself as an element and therefore does not belong to S (since by definition a set belongs to S only if it does not contain itself as an element). Hence $S \notin S$, and we have a contradiction. We conclude that the hypothesis must be wrong.
- 2 If we suppose $S \notin S$, then S does not contain itself as an element and therefore belongs to S (since by definition a set belongs to S if it does not contain itself as an element). Hence $S \in S$ and we have another contradiction. We conclude that this hypothesis must be wrong too!

Russell's paradox: The set of all sets that do not belong to themselves belongs to itself if and only if it does not belong to itself.

Zermelo-Fraenkel axioms

- ▶ The formal definition of the concept of set is based on the **Zermelo-Fraenkel system of axioms**, abbreviated as **ZF**. This system of axioms includes the standard axioms of axiomatic set theory on which, together with the *axiom of choice*, all of ordinary mathematics is based.
- ▶ The **axiom of regularity** states that “Every non-empty set A contains an element disjoint from A ”.
- ▶ The **axiom of pairing** states that “Given two objects, there exists a set whose elements are the two objects”

Observation 7

No set is an element of itself

Proof

Given a set A , we apply the axiom of regularity to $\{A\}$, which is a set by the axiom of pairing. We thus obtain the set $\{A, A\}$, which we abbreviate to $\{A\}$ since sets cannot contain repeated objects (it is a special case of a pair).

By the axiom of regularity there must exist an element of $\{A\}$ disjoint from $\{A\}$. Since the only element of $\{A\}$ is A , A is disjoint from $\{A\}$.

Hence, since $A \cap \{A\} = \emptyset$, we cannot have $A \in A$ (by the definition of disjoint). □

- ▶ The other ZF axioms are not part of the basic course in Mathematical Analysis.

A further proof of $0,\overline{9} = 1$

- ▶ A further proof of

$$0,\overline{9} = 1$$

starts from the assumption that two numbers are equal if and only if their difference is equal to zero, and it is based on computing the value of $1 - 0,\overline{9}$.

- ▶ This proof is based on the fact that 0 is the only non-negative number less than all the reciprocals of the positive integers, or equivalently that there is no number greater than every integer. This is the **Archimedean property**, which holds for the rational and the real numbers.

Proof (1/2)

We write the number $0,999\dots$ with n digits after the decimal point as $0,(9)_n$, hence $0,(9)_1 = 0.9$, $0,(9)_2 = 0.99$, $0,(9)_3 = 0.999$, and so on.

Given $\frac{1}{10^n} = 0,0\dots01$, with n digits after the decimal point, the addition rules for decimal numbers imply

$$0,(9)_n + \frac{1}{10^n} = 1 \quad \text{moreover} \quad 0,(9)_n < 1, \forall n \in \mathbb{N}.$$

We must prove that 1 is the smallest number that is not less than all the $0,(9)_n$. For this it is enough to prove that, if a number x is not greater than 1 and not less than all the $0,(9)_n$, then $x = 1$.

Proof (2/2)

So let x be such that

$$0, (9)_n \leq x \leq 1$$

for every positive integer n . Hence

$$1 - 1 \leq 1 - x \leq 1 - 0, (9)_n$$

which, using basic arithmetic and the first equality established above, simplifies to

$$0 \leq 1 - x \leq \frac{1}{10^n}$$

This implies that the difference between 1 and x is less than the reciprocal of any positive integer. Hence this difference must be zero, and therefore $x = 1$; which in turn implies

$$0.999 \dots = 1$$



Summary

- ▶ **Russell's paradox**: the set of the sets that do not belong to themselves leads to a contradiction
- ▶ Axiomatic theory (ZF) avoids paradoxes: axiom of regularity, axiom of pairing
- ▶ No set is an element of itself
- ▶ $0, \overline{9} = 1$ also through the **Archimedean property**



Summary (1/2)

Sets and elements:

- ▶ Set, element and membership; definition by listing or by a property
- ▶ Number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ through decimal expansions

Relations between sets:

- ▶ Equality, inclusion; difference between $\in$ and $\subseteq$
- ▶ Empty set, cardinality, power set (2^n elements)

Summary (2/2)

Operations on sets:

- ▶ Intersection, union, difference, complement, Cartesian product
- ▶ Venn diagrams; commutative, associative, distributive properties, De Morgan's laws
- ▶ Properties of cardinality

Further topics:

- ▶ Russell's paradox and the Zermelo-Fraenkel axioms